



Study Abroad Intelligence Dashboard for Indian Students

Data Analytics Project Report

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Tech Stack: Microsoft Excel • Python (Pandas) • PostgreSQL • Power BI

Project Overview

The **Study Abroad Intelligence Dashboard for Indian Students** is an end-to-end **Business Intelligence and Data Analytics** project designed to help students compare international study destinations using data-driven insights. The project integrates educational, financial, career, and visa-related information into a single interactive dashboard, enabling informed decision-making based on affordability, employability, visa opportunities, and return on investment (ROI).

The project follows the complete data analytics lifecycle, including data collection, preprocessing, exploratory data analysis (EDA), SQL-based analysis, KPI development, and dashboard visualization. Using **Microsoft Excel, Python, PostgreSQL, and Power BI**, raw data is transformed into meaningful insights through interactive reports and visualizations.

The dashboard consists of four analytical pages, each focusing on a specific aspect of study abroad planning:

- **Executive Overview:** Presents key performance indicators (KPIs) such as total countries, total students, average tuition fee, average graduate salary, and ROI, along with country-wise comparisons.
- **Career & ROI Analysis:** Analyzes graduate salaries, employment rates, and ROI to evaluate career prospects across countries.
- **Visa & PR Analysis:** Compares visa acceptance rates, rejection rates, post-study work visa duration, and permanent residency (PR) availability.
- **Student Decision Hub:** Provides personalized country recommendations based on user-selected criteria such as budget, expected salary, and visa acceptance rate.

The dashboard includes interactive filters and dynamic visualizations that allow users to compare **countries** across different **years** and evaluation criteria. Overall, the project demonstrates practical skills in **data preprocessing, SQL querying, exploratory data analysis (EDA), dashboard development, business intelligence, and data storytelling** while supporting informed study abroad decisions for Indian students.

Business Problem

Every year, thousands of Indian students aspire to pursue higher education abroad. However, selecting the most suitable study destination is a challenging task due to the need to evaluate multiple factors such as tuition fees, living expenses, visa acceptance rates, post-study work opportunities, graduate salaries, employment rates, scholarships, and return on investment (ROI). This information is often scattered across university websites, government portals, and educational platforms, making comparison difficult and time-consuming.

As a result, students frequently rely on incomplete information or personal opinions rather than objective data, which may lead to higher educational costs, limited career opportunities, or poor return on investment.

To address this challenge, the **Study Abroad Intelligence Dashboard for Indian Students** was developed as a centralized data analytics platform. It integrates financial, academic, career, and visa-related information into a single interactive dashboard, enabling users to compare study destinations through KPIs and visualizations. The platform supports data-driven decision-making by providing comprehensive insights into affordability, career prospects, visa success rates, post-study work opportunities, and ROI.

Objective

The primary objective of the **Study Abroad Intelligence Dashboard for Indian Students** is to develop an end-to-end data analytics solution that enables students to compare international study destinations using interactive visualizations and data-driven insights. The project integrates financial, academic, career, and visa-related information into a single platform to support informed decision-making for students planning higher education abroad.

Specific Objectives

- Collect, clean, and preprocess study abroad data to ensure accuracy and consistency.
- Analyze key factors such as tuition fees, living expenses, graduate salaries, employment rates, visa acceptance rates, scholarships, and return on investment (ROI).
- Perform exploratory data analysis (EDA) using Python to identify meaningful trends and patterns.
- Use PostgreSQL to query and analyze data for answering business-related questions.
- Develop an interactive Power BI dashboard with KPI cards, filters, and visualizations for country comparison.
- Build a **Student Decision Hub** that recommends suitable study destinations based on user preferences such as budget, expected salary, and visa acceptance rate.
- Present actionable insights that assist Indian students in making data-driven study abroad decisions.
- Demonstrate practical skills in Excel, Python, PostgreSQL, Power BI, data visualization, and business intelligence through a real-world analytics project.

Dataset Description

The **Study Abroad Intelligence Dashboard** is developed using a structured dataset containing educational, financial, visa, and career-related information for **15 popular study abroad destinations** over the period **2016–2025**. The dataset consists of **150 records** and **21 features**, providing comprehensive insights into the key factors influencing Indian students' study abroad decisions.

The dataset includes information such as tuition fees, living expenses, graduate salaries, employment rates, visa acceptance rates, post-study work visa duration, permanent residency availability, scholarship opportunities, university rankings, and economic indicators. After preprocessing using **Microsoft Excel** and **Python**, the cleaned dataset was stored in **PostgreSQL** for analysis and visualized in **Power BI** to create an interactive dashboard that supports data-driven decision-making

Dataset Summary :

Attribute	Details
Dataset Name	Study Abroad Intelligence Dataset
Domain	Higher Education Analytics
File Format	CSV
Time Period	2016–2025
Number of Records	150
Number of Countries	15
Number of Features	21
Data Type	Structured Tabular Data
Tools Used	Microsoft Excel, SQL, Python, Power BI

Technology Stack

The **Study Abroad Intelligence Dashboard for Indian Students** was developed using industry-standard tools and technologies to perform data preprocessing, analysis, visualization, and dashboard development. Each technology contributed to a specific stage of the data analytics workflow.

Technology	Purpose
Microsoft Excel	Initial data validation
PostgreSQL (SQL)	Data storage, querying, filtering aggregation, and analytical reporting.
Python	Data preprocessing and Exploratory Data Analysis (EDA).
Pandas	Data loading, exploration, filtering, and manipulation.
Power BI	Interactive dashboard development and business intelligence reporting.
Power Query	Data import, transformation, and preparation before visualization.
DAX (Data Analysis Expressions)	Creating calculated measures, KPIs, and custom metrics in Power BI.
Git & GitHub	Version control, project documentation, and portfolio hosting.

Skills Demonstrated

- Data Analysis
- SQL Query Writing
- Exploratory Data Analysis (EDA)
- Data Visualization
- Dashboard Development
- KPI Design
- Business Intelligence
- Data Storytelling
- Analytical Thinking
- Version Control with Git & GitHub

Project Workflow

The project followed a structured end-to-end data analytics workflow to transform raw study abroad data into an interactive Business Intelligence dashboard. Each stage contributed to data preparation, analysis, visualization, and insight generation.

Step 1: Data Collection

- Imported the Study Abroad Intelligence Dataset.
- Reviewed the dataset structure and verified data consistency.

Step 2: Data Validation & Preprocessing (Microsoft Excel)

- Validated data accuracy and consistency.
- Reviewed data types and identified potential quality issues.
- Prepared the dataset for further analysis.

Step 3: Data Analysis (PostgreSQL)

- Imported the dataset into PostgreSQL.
- Executed SQL queries to analyze tuition fees, living expenses, graduate salaries, employment rates, visa acceptance rates, scholarships, and ROI.
- Generated aggregated insights for business analysis.

Step 4: Exploratory Data Analysis (Python & Pandas)

- Loaded the dataset using Pandas.
- Performed data preprocessing and exploratory data analysis (EDA).
- Identified trends, distributions, and relationships among key variables.

Step 5: Dashboard Development (Power BI)

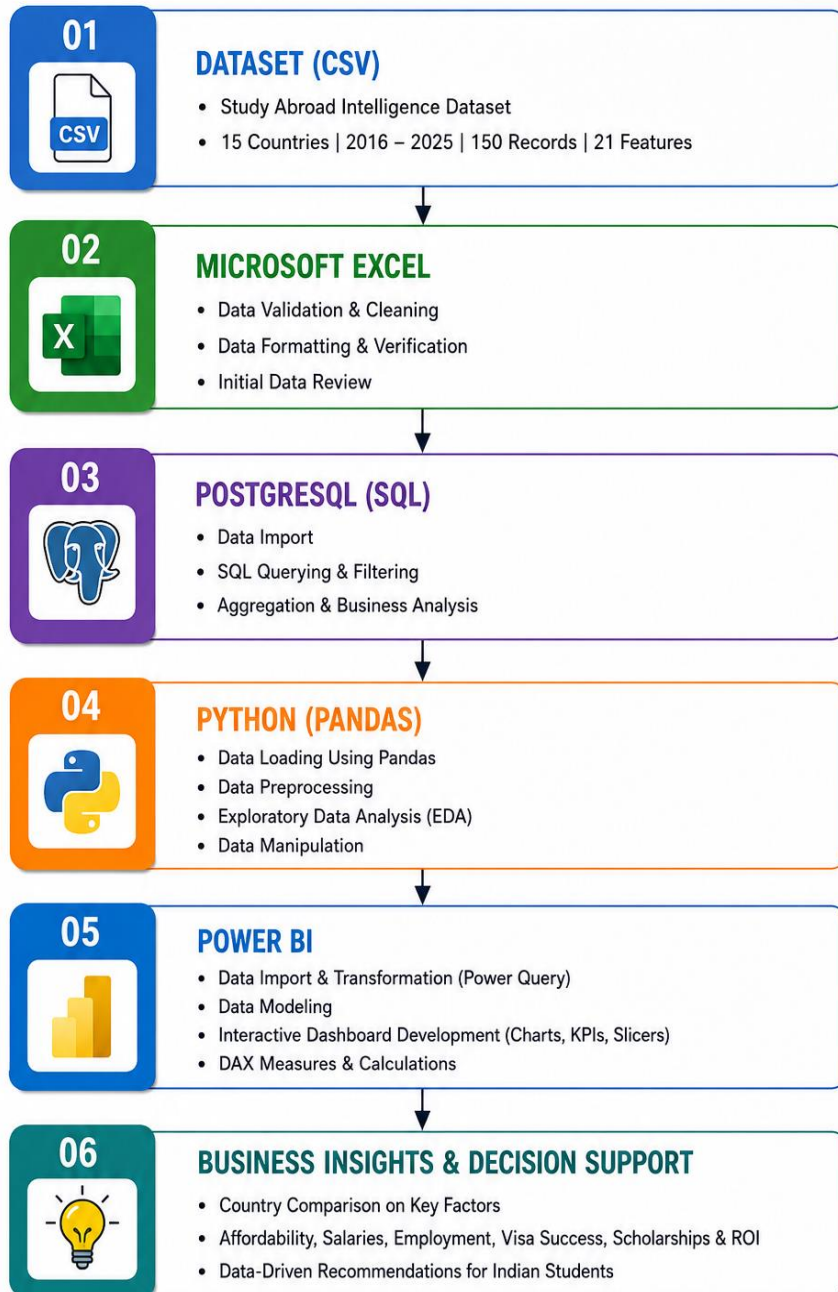
- Imported the processed dataset into Power BI.
- Developed interactive dashboards using KPI cards, charts, tables, and slicers.
- Created DAX measures and Power Query transformations to enhance reporting.

Step 6: Insight Generation

- Compared study destinations based on affordability, career prospects, visa success rates, scholarships, and ROI.
- Presented data-driven insights to support informed study abroad decisions

PROJECT METHODOLOGY WORKFLOW

End-to-End Workflow of Study Abroad Intelligence Dashboard



Excel Analysis

Overview

Microsoft Excel was used as the initial tool to review and validate the Study Abroad Intelligence Dataset before performing advanced analysis in PostgreSQL, Python, and Power BI. Since the dataset was already clean and structured, Excel was primarily used for data exploration, verification, and preliminary analysis.

Objectives

The objectives of using Microsoft Excel were:

- Review the dataset structure and attributes.
- Verify the completeness and consistency of the dataset.
- Explore country-wise and year-wise study abroad information.
- Perform preliminary analysis before importing the dataset into SQL and Power BI.

Activities Performed

The following tasks were carried out in Microsoft Excel:

- Imported and reviewed the CSV dataset.
- Verified column names and data formats.
- Applied sorting and filtering for exploratory analysis.
- Created Pivot Tables to summarize country-wise student data.
- Performed initial analysis of study abroad trends across different countries and years.

Year	Country	Number_of_Indian_Students	Tuition_Fee_USD	Living_Cost_USD	Average_Rent_USD	Transportation_Cost_USD	Total_Annual_Cost_USD	Student_Visa_Applications	Student_Visa_Approvals	Student_Visa_Rejections
2016	Canada	80006	24024	13024	6157	785	37048	89265	76747	12518
2017	Canada	81896	19031	18746	6111	1419	37777	89807	80909	8898
2018	Canada	87253	12078	17449	4084	1235	29527	98611	82552	16059
2019	Canada	92363	29700	11300	3639	593	41000	102150	90877	11273
2020	Canada	91405	23496	7339	5303	1593	30835	101897	71394	30503
2021	Canada	95723	10337	11913	11792	2294	22250	106571	76295	30276
2022	Canada	101394	34077	18087	8430	949	52164	116567	80767	35800
2023	Canada	100847	36674	13238	11564	2262	49912	115219	99190	16029
2024	Canada	108478	23261	21403	5427	1539	44664	123517	113618	9899
2025	Canada	107997	38021	5638	4332	2402	43659	115253	110118	5135
2016	United States	100629	40911	9120	4603	1851	50031	108859	89312	19547
2017	United States	110400	34711	21959	9897	747	56670	116165	77812	38353
2018	United States	106867	27654	7321	10923	987	34975	112873	90352	22521
2019	United States	117159	29219	18879	9235	1456	48098	132658	88455	44203
2020	United States	119126	15186	18824	5506	1070	34010	128672	118518	10154
2021	United States	125921	16899	18317	10456	1485	35216	131938	102162	29776
2022	United States	127939	42422	20947	5036	888	63369	135432	128701	6731
2023	United States	129055	9727	21640	3812	2243	31367	134753	112446	22307
2024	United States	133363	33474	15366	6772	918	48840	147641	110346	37295
2025	United States	136066	12552	7400	11308	936	19952	147735	102093	45642
2016	United Kingdom	84037	40664	5256	11586	1113	45920	95896	65411	30485
2017	United Kingdom	80053	28471	11671	6825	1535	40142	86738	80723	6015
2018	United Kingdom	83011	34956	19014	11689	519	53970	85878	80980	4898
2019	United Kingdom	85925	8614	16724	5941	1896	25338	90430	61661	28769
2020	United Kingdom	92056	27770	18491	6565	1046	46261	95802	83098	12704
2021	United Kingdom	95715	20609	5775	5664	1316	26384	101713	93367	8346
2022	United Kingdom	99185	8507	8552	9619	1207	17059	110562	82616	27946
2023	United Kingdom	97102	40073	11456	8467	1383	51529	99927	85528	14399
2024	United Kingdom	98445	14341	11286	9388	1861	25627	112448	96701	15747

Student_Visa_Rejections	Visa_Acceptance_Rate	Average_Graduate_Salary_USD	Graduate_Employment_Rate	Post_Study_Work_Visa_Months	PR_Availability	Number_of_Top_Universities	Average_QS_Ranking	Scholarships_Available
12518	85.98	86469	85.3	48	Yes	3	67	60
8898	90.09	28425	91.2	24	No	23	162	44
16059	83.71	80895	71.2	48	Yes	26	60	146
11273	88.96	33229	93.9	12	No	19	252	98
30503	70.06	52867	77.6	24	No	55	48	63
30276	71.59	60717	81.1	48	Yes	18	91	68
35800	69.29	60343	72.5	12	Yes	11	341	45
16029	86.09	35507	89.1	36	No	9	170	116
9899	91.99	47558	93.6	24	Yes	25	102	143
5135	95.54	80872	78.6	12	Yes	58	310	25
19547	82.04	45370	84.8	48	Yes	36	122	84
38353	66.98	29378	86.5	24	Yes	2	56	20
22521	80.05	63339	73.7	48	Yes	52	262	109
44203	66.68	70824	97.6	12	Yes	27	193	32
10154	92.11	32940	82.4	12	Yes	43	296	8
29776	77.43	31842	74.6	12	No	18	252	78
6731	95.03	65957	90.6	12	No	5	45	127
22307	83.45	72250	94.1	48	Yes	58	311	68
37295	74.74	45407	81.1	36	No	22	57	7
45642	69.11	50872	94.7	24	No	20	100	117
30485	68.21	85548	73.8	12	Yes	49	303	44
6015	93.07	83450	71.4	48	No	4	21	90
4898	94.3	73286	95.3	12	No	39	302	42
28769	68.19	79125	85.7	48	Yes	17	103	50
12704	86.74	53070	94.4	48	Yes	14	255	94
8346	91.79	78680	77.8	48	No	3	79	71
27946	74.72	67728	97.6	12	No	59	315	53
14399	85.59	49639	87.4	12	No	34	178	109
15747	86	87166	74.9	36	No	37	20	82

Row Labels	Sum of Number_of_Indian_Students
Australia	1101370
Canada	947362
France	1275971
Germany	534802
Ireland	1033701
Italy	1266288
Japan	236521
Netherlands	55431
New Zealand	208575
Russia	377446
Singapore	834807
South Korea	1128981
United Arab Emirates	830459
United Kingdom	920625
United States	1206525

Key Findings

The preliminary analysis in Excel revealed:

- The dataset contains 15 countries, 150 records, and 21 features covering the period 2016–2025.
- Country-wise student distribution was analyzed using Pivot Tables.
- Tuition fees, living costs, visa statistics, salaries, and employment data were available for every country and year.
- The dataset was well-structured and ready for further analysis.

Role of Excel in the Project

Microsoft Excel served as the first stage of the analytics workflow. It helped in understanding the dataset, validating the data, and performing basic exploratory analysis before moving to SQL, Python, and Power BI for advanced analytics and dashboard development.

Deliverables

The Excel stage helped in:

- Understanding the dataset structure.
 - Verifying data quality and consistency.
 - Performing preliminary exploratory analysis.
 - Generating Pivot Table summaries.
 - Preparing the dataset for SQL analysis and dashboard development.
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SQL Analysis

Overview

PostgreSQL was used to analyze the Study Abroad Intelligence Dataset by executing SQL queries for filtering, aggregation, ranking, and comparative analysis. The insights generated through SQL formed the basis for KPIs and visualizations used in the Power BI dashboard.

Objectives

The objectives of using SQL in this project were:

- Import the dataset into PostgreSQL.
- Perform analytical SQL queries.
- Compare countries using financial, academic, and career metrics.
- Generate aggregated reports for dashboard development.
- Prepare data for Power BI visualization.

SQL Environment

Attribute	Details
Database	PostgreSQL
Dataset Format	CSV
Number of Records	150
Number of Features	21
SQL Language	PostgreSQL SQL

SQL Concepts Used

The following SQL concepts were applied during the analysis:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- Aggregate Functions (SUM, AVG, COUNT, MAX, MIN)
- CASE Statements
- Aliases
- Filtering and Sorting

Query 1 – Total Number of Countries

Objective

Determine the total number of unique countries included in the dataset.

SQL Query

```
-- 1. Total Number of Countries.  
SELECT COUNT(DISTINCT "Country") AS total_countries  
FROM data;
```

Output

	total_countries 
1	15

Query 2 – Average Annual Study Cost by Country

Objective

Calculate the average annual study cost for each country to compare overall affordability.

SQL Query

```
-- 2. Average Annual Study Cost For Each Country.  
SELECT "Country", Round(AVG("Total_Annual_Cost_USD"), 2) AS avg_annual_cost  
FROM data  
GROUP BY "Country"  
ORDER BY Avg_Annual_Cost;
```

Output

	Country text	avg_annual_cost numeric
1	Japan	32159.80
2	Singapore	35140.80
3	New Zealand	35474.20
4	Germany	36061.20
5	United Kingdom	37958.40
6	Canada	38883.60
7	France	39606.90
8	Ireland	39710.50
9	Russia	40093.20
10	Australia	40275.30
11	Italy	41595.50
12	Netherlands	41834.10
13	United States	42252.80
14	South Korea	42482.70
15	United Arab Emirat...	44498.80

Query 3 – Top 5 Countries with the Lowest Annual Study Cost

Objective

Identify the five most affordable countries based on the average annual study cost.

SQL Query

```
-- 3. Top 5 Countries with the Lowest Annual Study Cost.
SELECT "Country", ROUND(AVG("Total_Annual_Cost_USD"), 2) AS avg_annual_cost
FROM data
GROUP BY "Country"
ORDER BY avg_annual_cost LIMIT 5;
```

Output

	Country text	avg_annual_cost numeric
1	Japan	32159.80
2	Singapore	35140.80
3	New Zealand	35474.20
4	Germany	36061.20
5	United Kingdo...	37958.40

Query 4 – Average Graduate Salary by Country

Objective

Compare the average graduate salary across different countries to evaluate career opportunities.

SQL Query

```
-- 4. Average Graduate Salary For Each Country.  
SELECT "Country", ROUND(AVG("Average_Graduate_Salary_USD"),2) AS avg_salary  
FROM data  
GROUP BY "Country" ORDER BY avg_salary;
```

Output

	Country text	avg_salary numeric
1	United States	50817.90
2	Netherlands	54204.50
3	New Zealand	54510.10
4	South Korea	54698.30
5	United Arab Emirat...	55909.50
6	Canada	56688.20
7	France	58162.50
8	Singapore	58814.00
9	Japan	58851.90
10	Italy	59107.60
11	Australia	59386.40
12	Germany	61418.10
13	Russia	62553.40
14	Ireland	65087.70
15	United Kingdom	73771.00

Query 5 - Top 5 Countries with the Highest Graduate Salaries

Objective

Identify the top five countries offering the highest average graduate salaries.

SQL Query

```
-- 5. Top 5 Countries with the Highest Graduate Salaries.
SELECT "Country", ROUND(AVG("Average_Graduate_Salary_USD"),2) AS avg_salary
FROM data
GROUP BY "Country" ORDER BY avg_salary DESC LIMIT 5;
```

Output

	Country text	avg_salary numeric
1	United Kingdo...	73771.00
2	Ireland	65087.70
3	Russia	62553.40
4	Germany	61418.10
5	Australia	59386.40

Query 6 – Average ROI by Country

Objective

Calculate the average Return on Investment (ROI) for each country to assess the financial benefits of studying abroad.

SQL Query

```
-- 6. Average ROI for Each Country.
SELECT
    "Country",
    ROUND(AVG("ROI")::numeric, 2) AS avg_roi
FROM data
GROUP BY "Country"
ORDER BY avg_roi;
```

Output

	Country text	avg_roi numeric
1	United Arab Emirat...	1.31
2	United States	1.34
3	Netherlands	1.36
4	South Korea	1.39
5	Italy	1.48
6	France	1.50
7	Canada	1.59
8	Russia	1.69
9	New Zealand	1.73
10	Ireland	1.74
11	Germany	1.76
12	Singapore	1.78
13	Australia	1.82
14	Japan	2.04
15	United Kingdom	2.26

Query 7 – Rank Countries Based on ROI

Objective

Rank all countries according to their average ROI to identify the best study destinations from an investment perspective.

SQL Query

```
-- 7. Rank countries based on ROI.
SELECT "Country", Round(AVG("ROI")::numeric, 2) AS avg_roi,
RANK() OVER (ORDER BY AVG("ROI") DESC) AS roi_rank
FROM data
GROUP BY "Country";
```

Output

	Country text	avg_roi numeric	roi_rank bigint
1	United Kingdom	2.26	1
2	Japan	2.04	2
3	Australia	1.82	3
4	Singapore	1.78	4
5	Germany	1.76	5
6	Ireland	1.74	6
7	New Zealand	1.73	7
8	Russia	1.69	8
9	Canada	1.59	9
10	France	1.50	10
11	Italy	1.48	11
12	South Korea	1.39	12
13	Netherlands	1.36	13
14	United States	1.34	14
15	United Arab Emirat...	1.31	15

Query 8 – Visa Success Rate by Country

Objective

Analyze the average student visa success rate for each country.

SQL Query

```
-- 8. VISA Success Rate for Each Country.  
SELECT "Country", ROUND(AVG("Visa_Success")::numeric,2) AS visa_success_rate  
FROM data  
GROUP BY "Country" ORDER BY visa_success_rate DESC;
```

Output

	Country text	visa_success_rate numeric
1	South Korea	84.29
2	Canada	83.33
3	France	83.19
4	Japan	83.17
5	Ireland	83.00
6	United Kingdom	82.95
7	Italy	82.31
8	Australia	82.28
9	Netherlands	82.14
10	Singapore	81.30
11	New Zealand	80.39
12	Germany	79.94
13	United States	78.76
14	Russia	77.85
15	United Arab Emirat...	77.64

Query 9 – Ranking Countries Based on Scholarship Opportunities

Objective

Rank countries according to the average scholarship opportunities available for international students.

SQL Query

```
-- 9. Ranking Countries Based on Scholarship Opportunities.
SELECT "Country", ROUND(AVG("Scholarships_Available")::numeric,2)
AS avg_scholarships,
RANK() OVER(ORDER BY AVG("Scholarships_Available") DESC)
AS scholarship_rank
FROM data
GROUP BY "Country";
```

Output

	Country text	avg_scholarships numeric	scholarship_rank bigint
1	Netherlands	99.90	1
2	New Zealand	81.70	2
3	Ireland	81.40	3
4	Canada	80.80	4
5	South Korea	78.80	5
6	Italy	76.60	6
7	Russia	76.20	7
8	Singapore	75.40	8
9	United Kingdom	72.50	9
10	Australia	66.80	10
11	France	65.60	11
12	United States	65.00	12
13	Germany	60.60	13
14	United Arab Emirat...	53.40	14
15	Japan	51.80	15

Query 10 – Top 5 Countries with the Highest Visa Success Rate

Objective

Identify the five countries with the highest average student visa success rates.

SQL Query

```
-- 10. Countries with the Highest Visa Success Rates.  
SELECT "Country", ROUND(AVG("Visa_Success")::numeric,2) AS visa_success_rate  
FROM data  
GROUP BY "Country" ORDER BY visa_success_rate DESC LIMIT 5;
```

Output

	Country text	visa_success_rate numeric
1	South Korea	84.29
2	Canada	83.33
3	France	83.19
4	Japan	83.17
5	Ireland	83.00

Query 11 – Highest Graduate Salary by Country

Objective

Identify the country offering the highest average graduate salary after completing higher education.

SQL Query

```
-- 11. Best Country for Students Seeking High Salaries after Graduation.  
SELECT "Country", ROUND(AVG("Average_Graduate_Salary_USD")::numeric,2)  
AS avg_salary  
FROM data  
GROUP BY "Country" ORDER BY avg_salary DESC LIMIT 1;
```

Output

	Country text	avg_salary numeric
1	United Kingdom	73771.00

Query 12 – Best Value for Money Countries

Objective

Identify countries that provide higher-than-average graduate salaries while maintaining lower-than-average annual study costs.

SQL Query

```
-- 12. Find Countries that offer the Best Value for Money
-- (Higher-than-average graduate salary and lower-than-average annual study cost)
SELECT
    "Country",
    ROUND(AVG("Average_Graduate_Salary_USD")::numeric, 2) AS avg_salary,
    ROUND(AVG("Total_Annual_Cost_USD")::numeric, 2) AS avg_cost
FROM data
GROUP BY "Country"
HAVING AVG("Average_Graduate_Salary_USD") >
    (SELECT AVG("Average_Graduate_Salary_USD") FROM data)
AND AVG("Total_Annual_Cost_USD") <
    (SELECT AVG("Total_Annual_Cost_USD") FROM data);
```

Output

	Country text	avg_salary numeric	avg_cost numeric
1	United Kingdom	73771.00	37958.40
2	Germany	61418.10	36061.20

Query 13 – Country Ranking Based on ROI & Affordability

Objective

Determine the best overall country by considering both Return on Investment (ROI) and Affordability Score.

SQL Query

```
-- 13. Best Country for International Students based on ROI and Affordability.
SELECT "Country", ROUND(AVG("ROI")::numeric,2) AS roi,
ROUND(AVG("Affordability")::numeric,2) AS affordability
FROM data
GROUP BY "Country" ORDER BY AVG("ROI") DESC,
AVG("Affordability") DESC LIMIT 1;
```

Output

	Country text	roi numeric	affordability numeric
1	United Kingdom	2.26	3.02

Query 14 – Country Summary Report

Objective

Generate a comprehensive country-wise summary report including annual study cost, graduate salary, ROI, visa success rate, scholarship availability, and affordability score.

SQL Query

```
-- 14. Country Summary Report:
SELECT
    "Country",
    ROUND(AVG("Total_Annual_Cost_USD")::numeric,2) AS average_cost,
    ROUND(AVG("Average_Graduate_Salary_USD")::numeric,2) AS average_salary,
    ROUND(AVG("ROI")::numeric,2) AS roi,
    ROUND(AVG("Visa_Success")::numeric,2) AS visa_success_rate,
    ROUND(AVG("Scholarships_Available")::numeric,2) AS scholarships,
    ROUND(AVG("Affordability")::numeric,2) AS affordability_score
FROM data
GROUP BY "Country"
ORDER BY "Country";
```

Output

	Country text	average_cost numeric	average_salary numeric	roi numeric	visa_success_rate numeric	scholarships numeric	affordability_score numeric
1	Australia	40275.30	59386.40	1.82	82.28	66.80	2.92
2	Canada	38883.60	56688.20	1.59	83.33	80.80	2.74
3	France	39606.90	58162.50	1.50	83.19	65.60	2.61
4	Germany	36061.20	61418.10	1.76	79.94	60.60	2.99
5	Ireland	39710.50	65087.70	1.74	83.00	81.40	2.64
6	Italy	41595.50	59107.60	1.48	82.31	76.60	2.54
7	Japan	32159.80	58851.90	2.04	83.17	51.80	3.54
8	Netherlands	41834.10	54204.50	1.36	82.14	99.90	2.57
9	New Zealand	35474.20	54510.10	1.73	80.39	81.70	3.10
10	Russia	40093.20	62553.40	1.69	77.85	76.20	2.85
11	Singapore	35140.80	58814.00	1.78	81.30	75.40	3.04
12	South Korea	42482.70	54698.30	1.39	84.29	78.80	2.65
13	United Arab Emirates	44498.80	55909.50	1.31	77.64	53.40	2.37
14	United Kingdom	37958.40	73771.00	2.26	82.95	72.50	3.02
15	United States	42252.80	50817.90	1.34	78.76	65.00	2.63

Key SQL Insights

The SQL analysis revealed several important findings:

- Countries offering higher graduate salaries generally had higher tuition fees.
- Visa acceptance rates varied significantly across destinations.
- Countries with better employment rates tended to provide stronger ROI.
- Scholarship availability improved overall affordability.
- ROI analysis helped identify destinations offering better long-term financial value.

Role of SQL in the Project

SQL was used to retrieve, aggregate, and analyze study abroad data. The query results were used to build KPIs and interactive visualizations in Power BI, enabling meaningful business insights and country comparisons.

Deliverables

- Country-wise analytical reports
- Financial and career comparisons
- Visa and employment analysis
- KPI-ready datasets
- Business insights for Power BI dashboards

Python Analysis

Overview

Python was used to preprocess and prepare the Study Abroad Intelligence dataset before performing SQL analysis and developing the Power BI dashboard. Using the Pandas library, the dataset was inspected, validated, and enhanced through feature engineering. The processed dataset was then loaded into PostgreSQL for further analysis.

Objectives

The objectives of using Python in this project were:

- Load the dataset using Pandas.
- Explore the dataset structure.
- Generate descriptive statistics.
- Verify missing and duplicate values.
- Create additional analytical features.
- Export the processed dataset.
- Load the processed dataset into PostgreSQL

Python Environment

Attribute	Details
Programming Language	Python 3.x
Library	Pandas
Database Connector	SQLAlchemy
Database	PostgreSQL
IDE	Jupyter Notebook

Python Analysis Performed

Task 1 - Load the Dataset

Objective

Import the Study Abroad CSV dataset into a Pandas Data Frame.

```
1. Importing Pandas Library

import pandas as pd

[1] ✓ 6.6s

2. Reading Study Abroad Dataset CSV File

df = pd.read_csv("study_abroad.csv")

[6] ✓ 0.0s
```

Task 2 – Dataset Information

Objective

Examine the dataset structure, column names, data types, and record count.

```
3. Checking DataFrame(Dataset) Information

df.info()

[5] ✓ 0.0s

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Year                                  150 non-null    int64
1   Country                              150 non-null    object
2   Number_of_Indian_Students            150 non-null    int64
3   Tuition_Fee_USD                      150 non-null    int64
4   Living_Cost_USD                      150 non-null    int64
5   Average_Rent_USD                     150 non-null    int64
6   Transportation_Cost_USD              150 non-null    int64
7   Total_Annual_Cost_USD                150 non-null    int64
8   Student_Visa_Applications            150 non-null    int64
9   Student_Visa_Approvals               150 non-null    int64
10  Student_Visa_Rejections              150 non-null    int64
11  Visa_Acceptance_Rate                 150 non-null    float64
12  Average_Graduate_Salary_USD          150 non-null    int64
13  Graduate_Employment_Rate             150 non-null    float64
14  Post_Study_Work_Visa_Months          150 non-null    int64
15  PR_Availability                      150 non-null    object
16  Number_of_Top_Universities           150 non-null    int64
17  Average_QS_Ranking                  150 non-null    int64
18  Scholarships_Available               150 non-null    int64
19  GDP_Per_Capita_USD                  150 non-null    int64
20  Inflation_Rate                      150 non-null    float64
dtypes: float64(3), int64(16), object(2)
memory usage: 24.7+ KB
```

Task 3 – Descriptive Statistics

Objective

Generate summary statistics for all numerical features.

4. Checking DataFrame Description

```
df.describe()
```

[7] ✓ 0.1s Python

	Year	Number_of_Indian_Students	Tuition_Fee_USD	Living_Cost_USD	Average_Rent_USD	Transportation_Cost_USD	Total_Annual_Cost_USD	Student_Visa_Application_Fee_USD
count	150.000000	150.000000	150.000000	150.000000	150.000000	150.000000	150.000000	150.000000
mean	2020.500000	79725.760000	25703.506667	13498.346667	7674.333333	1518.433333	39201.853333	86474.110000
std	2.881904	40590.307612	10414.351115	5100.139440	2716.212483	562.811881	11248.328850	44005.660000
min	2016.000000	4891.000000	6363.000000	5113.000000	2751.000000	519.000000	14855.000000	5269.000000
25%	2018.000000	41371.500000	15421.000000	8798.000000	5441.750000	989.250000	30854.000000	44938.000000
50%	2020.500000	90458.500000	27374.000000	13131.000000	7757.000000	1492.500000	39875.500000	96864.500000
75%	2023.000000	111116.250000	34200.000000	18424.250000	10103.000000	1981.250000	47238.750000	121305.500000
max	2025.000000	145641.000000	44521.000000	21959.000000	11933.000000	2497.000000	64984.000000	159659.000000

Task 4 – Missing Value Analysis

Objective

Check whether the dataset contains any missing values.

```
5. Checking for the Missing Values

df.isnull().sum()

[8] ✓ 0.0s

... Year 0
Country 0
Number_of_Indian_Students 0
Tuition_Fee_USD 0
Living_Cost_USD 0
Average_Rent_USD 0
Transportation_Cost_USD 0
Total_Annual_Cost_USD 0
Student_Visa_Applications 0
Student_Visa_Approvals 0
Student_Visa_Rejections 0
Visa_Acceptance_Rate 0
Average_Graduate_Salary_USD 0
Graduate_Employment_Rate 0
Post_Study_Work_Visa_Months 0
PR_Availability 0
Number_of_Top_Universities 0
Average_QS_Ranking 0
Scholarships_Available 0
GDP_Per_Capita_USD 0
Inflation_Rate 0
dtype: int64
```

Task 5 – Duplicate Record Analysis

Objective

Identify duplicate records in the dataset.

```
6. Checking For Duplicates Values

df.duplicated().sum()

[9] ✓ 0.0s

... np.int64(0)
```

Feature Engineering

Task 6 – ROI Feature Creation

Objective

Create a new **Return on Investment (ROI)** feature by comparing graduate salary with annual study cost.

```
3. Creating ROI Score Column

df["ROI"] = (df["Average_Graduate_Salary_USD"] / df["Total_Annual_Cost_USD"])

df.columns

Index(['Year', 'Country', 'Number_of_Indian_Students', 'Tuition_Fee_USD',
      'Living_Cost_USD', 'Average_Rent_USD', 'Transportation_Cost_USD',
      'Total_Annual_Cost_USD', 'Student_Visa_Applications',
      'Student_Visa_Approvals', 'Student_Visa_Rejections',
      'Visa_Acceptance_Rate', 'Average_Graduate_Salary_USD',
      'Graduate_Employment_Rate', 'Post_Study_Work_Visa_Months',
      'PR_Availability', 'Number_of_Top_Universities', 'Average_QS_Ranking',
      'Scholarships_Available', 'GDP_Per_Capita_USD', 'Inflation_Rate',
      'ROI'],
      dtype='object')
```

Task 7 – Visa Success Feature

Objective

Calculate the student visa success percentage using visa approvals and applications.

```
4. Creating Visa Success Score

df["Visa_Success"] = (df["Student_Visa_Approvals"] / df["Student_Visa_Applications"]) * 100

df.columns

Index(['Year', 'Country', 'Number_of_Indian_Students', 'Tuition_Fee_USD',
      'Living_Cost_USD', 'Average_Rent_USD', 'Transportation_Cost_USD',
      'Total_Annual_Cost_USD', 'Student_Visa_Applications',
      'Student_Visa_Approvals', 'Student_Visa_Rejections',
      'Visa_Acceptance_Rate', 'Average_Graduate_Salary_USD',
      'Graduate_Employment_Rate', 'Post_Study_Work_Visa_Months',
      'PR_Availability', 'Number_of_Top_Universities', 'Average_QS_Ranking',
      'Scholarships_Available', 'GDP_Per_Capita_USD', 'Inflation_Rate', 'ROI',
      'Visa_Success'],
      dtype='object')
```

Task 8 – Affordability Score

Objective

Generate an affordability score to compare countries based on annual study cost.

```
5. Creating Affordability Score Column

df["Affordability"] = (1/df["Total_Annual_Cost_USD"]) * 100000

df.columns

Index(['Year', 'Country', 'Number_of_Indian_Students', 'Tuition_Fee_USD',
      'Living_Cost_USD', 'Average_Rent_USD', 'Transportation_Cost_USD',
      'Total_Annual_Cost_USD', 'Student_Visa_Applications',
      'Student_Visa_Approvals', 'Student_Visa_Rejections',
      'Visa_Acceptance_Rate', 'Average_Graduate_Salary_USD',
      'Graduate_Employment_Rate', 'Post_Study_Work_Visa_Months',
      'PR_Availability', 'Number_of_Top_Universities', 'Average_QS_Ranking',
      'Scholarships_Available', 'GDP_Per_Capita_USD', 'Inflation_Rate', 'ROI',
      'Visa_Success', 'Affordability'],
      dtype='object')
```

Task 9 – Export Processed Dataset

Objective

Save the cleaned and feature-engineered dataset for further analysis.

```
6. Saving the Dataset After Applying Featured Engineering

df.to_csv("study_abroad_features.csv", index=False)

print("Feature Engineering Completed!")

Feature Engineering Completed!
```

Loading Dataset into PostgreSQL

Task 10 – Import Dataset into PostgreSQL

Objective

Load the processed dataset into PostgreSQL using SQLAlchemy to enable SQL-based analysis.

```
1. Import Libraries

import pandas as pd
from sqlalchemy import create_engine

2. Read CSV

df = pd.read_csv("study_abroad_features_latest.csv", encoding="latin1")

3. PostgreSQL Connection

engine = create_engine(
    "postgresql://postgres:admin@localhost:5432/Study_Abroad"
)

4. Load into PostgreSQL

df.to_sql(
    "data",          # table name
    engine,
    if_exists="replace",
    index=False
)
```

Role of Python in the Project

Python was used for data exploration, validation, and feature engineering. Using Pandas, additional analytical features were created, and the processed dataset was prepared for SQL analysis and Power BI dashboard development.

Deliverables

- Dataset inspection
- Summary statistics
- Missing value verification
- Duplicate record verification
- ROI calculation
- Visa Success Rate calculation
- Affordability Score
- Processed dataset export
- PostgreSQL-ready dataset

Power BI Dashboard

Overview

Power BI was used to develop an interactive Business Intelligence dashboard for analyzing study abroad data. The dashboard combines financial, academic, career, and visa-related information into dynamic visualizations, enabling Indian students to compare study destinations and make data-driven decisions.

Objectives

- The objectives of developing the Power BI dashboard were:
- Create an interactive dashboard for study abroad analysis.
- Visualize key performance indicators (KPIs).
- Compare countries across multiple factors.
- Support data-driven decision-making.
- Present insights in a clear and user-friendly format.

Power BI Features Used

Feature	Purpose
Power Query	Data transformation
DAX Measures	KPI calculations
Cards	Display KPIs
Bar & Column Charts	Country comparison
Line Charts	Trend analysis
Slicers	Interactive filtering
Buttons	Page navigation

Dashboard Pages

Page 1 - Introduction

Objective

Provide an overview of the project, its objectives, and the purpose of the dashboard.



Page 2 - Executive Overview

Objective

Present key performance indicators (KPIs) and a high-level summary of study abroad trends.

Key Metrics Displayed

- Total Countries
- Total Indian Students
- Average Annual Study Cost
- Average Graduate Salary
- Average ROI

- Average Visa Success Rate



Page 3 – Career & ROI Analytics

Objective

Analyze graduate salaries, employment opportunities, and Return on Investment (ROI) across countries.

Insights Included

- Graduate salary comparison
- ROI comparison
- Employment trends
- Top-performing countries



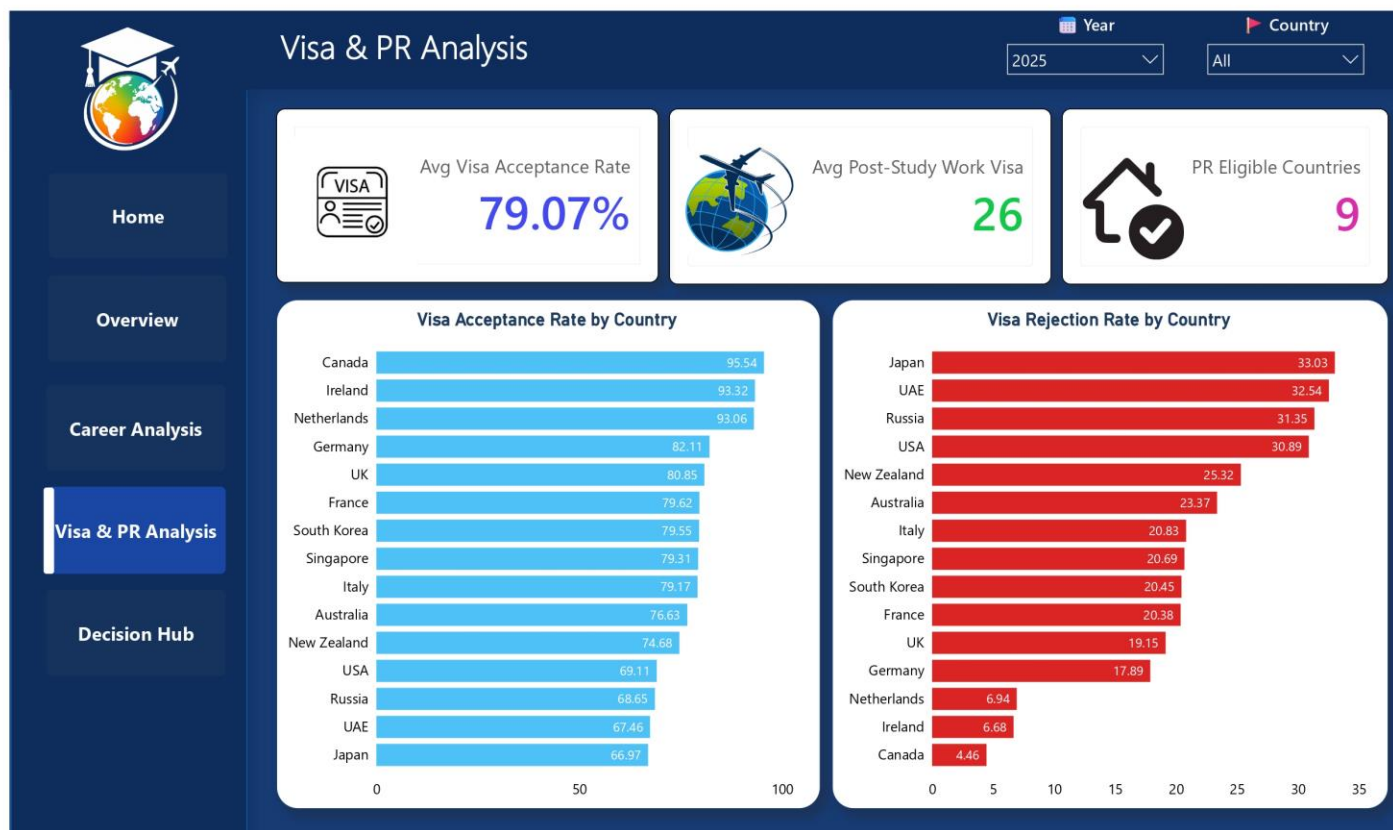
Page 4 – Visa & PR Analysis

Objective

Compare visa success rates, post-study work visa duration, and permanent residency (PR) opportunities.

Insights Included

- Visa success rate
- Student visa approvals
- Post-study work duration
- PR availability by country



Page 5 – Student Decision Hub

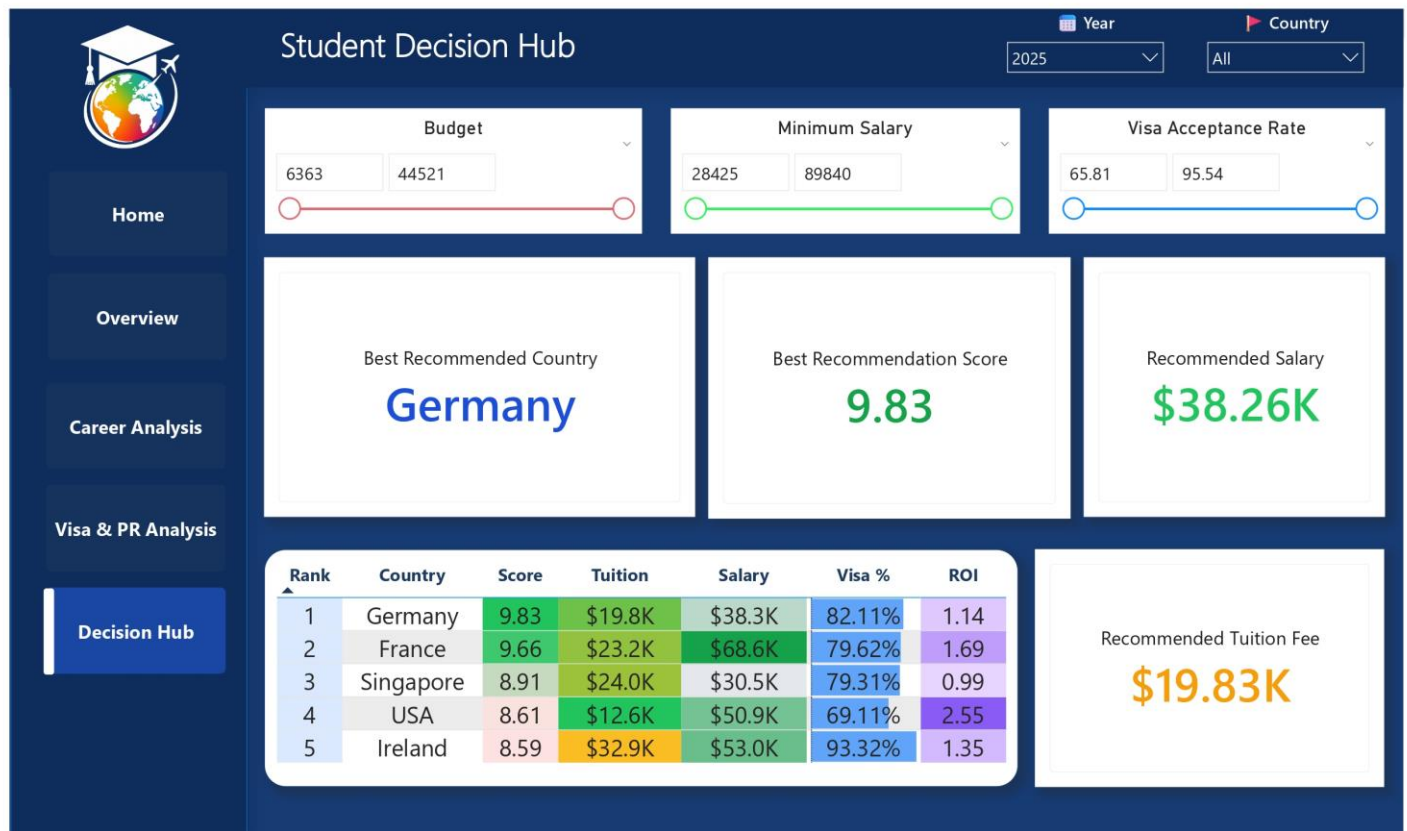
Objective

Provide a comprehensive comparison of study abroad destinations using affordability, scholarships, salary, ROI, and visa metrics to support informed decision-making.

Insights Included

- Country comparison
- Affordability score
- Scholarship opportunities
- ROI ranking

- Recommended study destinations



Dashboard Interactivity

The dashboard includes several interactive features to enhance user experience:

- Country slicer
- Year slicer
- Cross-filtering between visuals
- Interactive charts
- Drill-down capability
- Navigation buttons between report pages

Key KPIs

The dashboard presents the following KPIs:

- Total Countries
- Total Indian Students
- Average Annual Study Cost

- Average Graduate Salary
- Average Return on Investment (ROI)
- Average Visa Success Rate
- Average Employment Rate
- Scholarship Availability

Role of Power BI in the Project

Power BI served as the final visualization and reporting layer of the project. It integrated insights generated from PostgreSQL and Python into an interactive dashboard, enabling users to explore study abroad data through dynamic visualizations, KPI cards, filters, and comparative analysis. The dashboard supports data-driven decision-making by presenting complex information in a clear and user-friendly format.

Dashboard Deliverables

The Power BI dashboard provides:

- Executive overview of study abroad trends
- Country-wise comparison of tuition fees, salaries, and ROI
- Career and employment analysis
- Visa and permanent residency (PR) insights
- Scholarship and affordability analysis
- Student Decision Hub for destination comparison
- Interactive filtering, drill-down, and cross-filtering capabilities

Key Business Insights

The analysis of the Study Abroad Intelligence dataset revealed several valuable insights that can help Indian students make informed decisions when selecting an international study destination.

1. Significant Cost Differences Across Countries

The analysis revealed considerable variation in annual study costs across countries. Some destinations provide quality education at a comparatively lower cost, making them more affordable for Indian students.

2. Graduate Salaries Vary Across Countries

Graduate salaries differ significantly among destinations. While some countries offer higher earning potential, these often come with higher education and living costs, making ROI an important evaluation metric.

3. Return on Investment (ROI) is a Critical Decision Factor

Countries with higher ROI provide better long-term financial value by balancing study costs with expected graduate salaries.

4. Visa Success Rates Differ Between Countries

Student visa approval rates vary across destinations. Countries with consistently higher visa success rates may offer a smoother application process and lower uncertainty for prospective students.

5. Scholarship Opportunities Improve Affordability

Scholarship availability plays an important role in reducing the financial burden of studying abroad. Countries with greater scholarship opportunities become more accessible to students with limited budgets.

6. Affordability and Career Growth Must Be Balanced

Selecting a study destination should not be based solely on the lowest cost. A balanced evaluation of affordability, graduate salary, employment opportunities, and ROI leads to better long-term outcomes.

7. Interactive Dashboard Enables Better Decision-Making

The Power BI dashboard combines multiple indicators into a single interactive platform, allowing users to compare countries efficiently and make data-driven study abroad decisions.

Overall Insight

The analysis demonstrates that selecting an international study destination should be based on a combination of affordability, career prospects, visa opportunities, and long-term financial returns rather than a single factor. By integrating these indicators into an interactive dashboard, the project enables Indian students to make informed, data-driven decisions aligned with their academic and career goals.

Business Recommendations

Based on the analysis of the **Study Abroad Intelligence Dashboard**, the following recommendations are proposed to support Indian students, education consultants, and other stakeholders in making informed study abroad decisions.

1. Prioritize ROI Over Tuition Fees

Students should evaluate the **Return on Investment (ROI)** instead of selecting a destination solely based on lower tuition fees. A country with higher education costs may provide significantly better graduate salaries and long-term financial benefits.

2. Consider Visa Success Rates During Country Selection

Students should compare visa approval rates before choosing a study destination. Countries with higher visa success rates can reduce application uncertainty and improve the chances of successful admission.

3. Focus on Overall Affordability

Decision-making should include the total annual study cost, including tuition fees, living expenses, rent, and transportation, rather than tuition fees alone.

4. Explore Scholarship Opportunities

Students should prioritize countries that offer more scholarships, as financial aid can significantly reduce education expenses and improve affordability.

5. Evaluate Career Opportunities Along with Education

Graduate salary and employment rate should be considered alongside academic quality to ensure better career prospects after graduation.

6. Leverage Data-Driven Decision Making

Students and education consultants should use interactive dashboards to compare countries based on affordability, graduate salaries, ROI, scholarships, employment opportunities, and visa success rates instead of relying on isolated information sources.

7. Regularly Update the Dataset

Since tuition fees, living costs, visa policies, and graduate salaries change over time, the dataset and dashboard should be updated periodically to ensure accurate and reliable recommendations.

Challenges

During the development of the **Study Abroad Intelligence Dashboard**, several challenges were encountered while preparing the data, performing analysis, and designing the interactive dashboard. These challenges provided valuable learning experiences throughout the project.

1. Understanding the Dataset

Initially, understanding the relationships between different variables such as tuition fees, graduate salaries, visa acceptance rates, scholarships, and employment rates required careful exploration before meaningful analysis could be performed.

2. Feature Engineering

Additional business metrics such as **ROI (Return on Investment)**, **Visa Success Rate**, and **Affordability Score** were not available in the original dataset. These features had to be created using Python to support deeper analysis.

3. Importing Data into PostgreSQL

Loading the processed dataset into PostgreSQL required configuring the database connection and using SQLAlchemy. Ensuring that the data was imported correctly was essential before performing SQL analysis.

4. Writing Business-Oriented SQL Queries

Developing SQL queries that answered real business questions required the use of aggregate functions, grouping, ranking, filtering, and calculated metrics to generate meaningful insights.

5. Designing an Interactive Dashboard

Selecting appropriate visualizations and organizing multiple dashboard pages while maintaining clarity and usability was a key design challenge. The dashboard needed to present complex information in an easy-to-understand format.

6. Building Meaningful KPIs

Identifying the most relevant KPIs, such as average study cost, graduate salary, ROI, visa success rate, and affordability, was important to ensure that the dashboard provided actionable insights.

7. Integrating Multiple Tools

The project involved the integration of **Excel, Python, PostgreSQL, and Power BI**. Ensuring a smooth workflow between these tools while maintaining data consistency was an important part of the project.

Project Outcome

The **Study Abroad Intelligence Dashboard** was successfully developed as an end-to-end Data Analytics and Business Intelligence solution using **Microsoft Excel, Python, PostgreSQL, and Power BI**. The project transformed raw study abroad data into an interactive dashboard that enables Indian students to compare international study destinations based on affordability, career opportunities, visa success rates, scholarships, and Return on Investment (ROI).

Project Outcomes

1. End-to-End Data Analytics Solution

Successfully completed the entire data analytics lifecycle, including data validation, preprocessing, feature engineering, SQL analysis, and interactive dashboard development

2. Interactive Business Intelligence Dashboard

Developed a multi-page Power BI dashboard featuring KPI cards, interactive charts, slicers, filters, and drill-down capabilities for effective country comparison.

3. Data-Driven Insights

Generated meaningful insights into:

- Annual study costs
- Graduate salaries
- Return on Investment (ROI)
- Visa success rates
- Scholarship availability
- Employment opportunities
- Affordability

These insights support informed decision-making for Indian students planning to study abroad.

4. Enhanced Analytical Dataset

Created additional analytical features such as **ROI, Visa Success Rate, and Affordability Score**, enabling deeper analysis and more meaningful comparisons.

5. Business Analysis Using SQL

Answered real-world business questions through SQL queries, including country ranking, affordability comparison, salary analysis, and visa success evaluation.

6. Professional Dashboard Design

Designed a user-friendly dashboard that presents complex study abroad data through interactive visualizations, making it easier to compare countries and explore trends.

7. Practical Skill Development

Strengthened practical skills in:

- Microsoft Excel
- Python (Pandas)
- PostgreSQL (SQL)
- Power BI
- Data Visualization
- Feature Engineering
- Business Intelligence
- End-to-End Data Analytics

Future Enhancement

The current dashboard provides valuable insights into study abroad destinations using historical data. In the future, additional features and technologies can be incorporated to make the solution more intelligent, interactive, and suitable for real-world use.

1. Real-Time Data Integration

Integrate the dashboard with live data sources or APIs to automatically update tuition fees, living costs, visa policies, exchange rates, and university rankings.

2. University-Level Analysis

Expand the dataset to include individual universities, allowing students to compare institutions based on tuition fees, rankings, graduate salaries, placement rates, and scholarship opportunities.

3. AI-Based Country Recommendation System

Develop an AI-powered recommendation engine that suggests the most suitable study destinations based on a student's budget, preferred course, academic profile, and career goals.

4. Predictive Analytics

Use Machine Learning models to forecast future tuition fees, graduate salaries, visa approval trends, and employment opportunities based on historical data.

5. Personalized Student Dashboard

Create a personalized dashboard where users can enter their preferences and receive customized recommendations, estimated study costs, and expected ROI.

6. Web-Based Dashboard

Deploy the dashboard as a web application so that students and education consultants can access it from any device without requiring Power BI Desktop.

7. Additional Performance Metrics

Include more analytical metrics such as:

- Exchange Rate Impact
- Cost of Living Index
- University Ranking Trends
- Job Market Demand
- Student Satisfaction Score

- Graduate Placement Rate

8. Automated Data Refresh

Configure scheduled data refresh to keep the dashboard updated with the latest information without manual intervention.

Conclusion

The **Study Abroad Intelligence Dashboard** successfully demonstrates an end-to-end **Data Analytics and Business Intelligence** workflow by transforming raw study abroad data into meaningful insights. Using **Microsoft Excel, Python (Pandas), PostgreSQL, and Power BI**, the project analyses key factors such as tuition fees, living expenses, graduate salaries, visa success rates, scholarships, affordability, and Return on Investment (ROI).

The project integrates data preprocessing, feature engineering, SQL-based analysis, and interactive dashboard development to create a centralized platform for comparing international study destinations. Through dynamic visualizations, KPI cards, and interactive filters, the dashboard enables users to evaluate multiple factors simultaneously, supporting informed and data-driven decision-making.

Beyond its practical application, the project strengthened technical skills in data preprocessing, SQL querying, feature engineering, business intelligence, dashboard development, and data visualization. It also demonstrates the ability to implement a complete analytics pipeline while presenting insights in a clear, interactive, and professional manner.

Overall, the **Study Abroad Intelligence Dashboard** serves as a practical decision-support tool for Indian students planning to pursue higher education abroad. It showcases the effective application of modern data analytics techniques to solve a real-world problem and represents a portfolio-ready project that reflects analytical thinking, technical expertise, and business intelligence skills.